Macroscopic Dephasing in LMT Atom Interferometry from a Fixed-Cell Transport Model

A Short Letter on the NDK Momentum-Quadratic Contrast Floor

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Abstract

A central open question in macroscopic quantum mechanics is whether contrast loss in large-momentum-transfer (LMT) atom interferometry remains entirely reducible to environmental and technical noise, or whether a genuinely irreducible geometric floor emerges once the momentum separation and interrogation time become sufficiently large. In the Null-Directional Kinematics (NDK) framework, this question is sharpened by a finite-throughput transport picture of the directional vacuum. The present Letter develops a compact phenomenological bridge between near-term atom interferometers and the cosmological strain scale of that framework, which in the broader NDK program is calibrated against the high-redshift supermassive-black-hole collapse envelope derived in [1]. The bridge is built from two ingredients: a dimensionless transport-burden variable and a fixed pre-split coherence-cell ansatz. The resulting leading-order prediction is a residual dephasing law

$$D_{\mathrm{pred}} \equiv -\ln V_{\mathrm{res}} \propto T\, n^2 k_{\mathrm{eff}}^2,$$

with a coefficient controlled by the ratio of the interferometric coherence-support width to the pre-split coherence length of the atomic source. A baseline benchmark together with a broad sweep over coherence and geometric parameters is then used to compare current $n \sim 10^2$ platforms to the $n \sim 10^3$ regime relevant to MAGIS-like architectures, while the directional-strain threshold is held fixed at the representative benchmark value $\mathcal{I}_*/A \approx 3.28$ inferred from the SMBH-envelope analysis of [1]. Within the fixed-cell ansatz adopted here, the $n \sim 10^3$ regime enters the percent to few-percent dephasing window while the $n \sim 10^2$ regime remains parametrically lower. The experimental discriminator is therefore sharp: at fixed interrogation time, the fixed-cell NDK bridge predicts a residual contrast loss with leading quadratic dependence on LMT order.

1 Introduction

The empirical success of quantum mechanics rests on the linear superposition principle, yet the question of whether macroscopic coherence is fundamentally unlimited remains unresolved. In practice, visibility loss in matter-wave interferometry is modeled through environmental coupling, finite pulse fidelity, laser phase noise, wavefront effects, background gas scattering, expansion of the cloud, and other apparatus-specific systematics. At the same time, the continued development of large-momentum-transfer (LMT) atom interferometry pushes experiments into regimes where path separation, coherence time, and phase-space burden grow dramatically [2–4].

This makes long-baseline instruments especially interesting as discriminators of any residual dephasing law that is intrinsic rather than purely technical. MAGIS-100, for example, is designed as a 100 m-scale atom interferometer and as a pathfinder for precisely the large-T, large-LMT control problem relevant to new-physics searches and future long-baseline interferometers [4,5]. The present Letter asks whether such architectures should be expected to encounter a residual contrast floor whose leading dependence is not merely linear in pulse number or interrogation time, but instead scales with the geometric combination $T(\Delta p)^2$.

In the broader NDK framework, the directional vacuum is modeled as a finite-throughput informational substrate, and its transport sector is governed by Fisher information in the privileged amplitude representation $P = |\Psi|^2$ [6,7]. Earlier work identified a momentum-quadratic dephasing signature as a possible experimental consequence of this transport picture. The specific goal of the present Letter is narrower and more practical: to formulate a compact phenomenological bridge between that transport framework and LMT atom interferometry, and then to show what that bridge implies for current $n \sim 10^2$ instruments versus future $n \sim 10^3$ regimes.

The main result is a fixed-cell ansatz in which the transport burden grows kinematically with $Tn^2k_{\rm eff}^2$, while the operational coherence cell is fixed by the pre-split source coherence rather than by the post-splitting momentum burden itself. This separation preserves the momentum-quadratic signature and yields a direct experimental forecast: for a broad but physically plausible range of coherence parameters, MAGIS-like $n \sim 10^3$ architectures enter the 1% to few-% dephasing window, while current $n \sim 10^2$ platforms remain substantially lower.

2 The phenomenological bridge

2.1 Transport burden

The first ingredient is a dimensionless transport-burden variable,

$$\mathcal{D}_{\rm tr} \equiv \pi\,\chi\,\mathcal{I}_*/A\,c\,\ell_{\rm P}\,T\,n^2k_{\rm eff}^2, \tag{1}$$

where T is the interrogation time, n is the effective LMT order, $k_{\rm eff}$ is the effective wavevector, $\ell_{\rm P}$ is the Planck length, and χ is an order-unity geometric factor. The factor $\mathcal{I}_*/A$ is the cosmologically calibrated directional-strain threshold inherited from the broader NDK program [6], and is fixed phenomenologically in the SMBH-envelope analysis of [1] by the early-universe supermassive-black-hole collapse envelope, anchored to high-redshift quasar data such as [8], rather than by laboratory data. In the present Letter, this factor is not treated as a swept laboratory parameter, but is fixed to the representative benchmark value $\mathcal{I}_*/A \approx 3.28$ inferred from that cosmological calibration. This benchmark is used here as an adopted phenomenological input rather than as a claimed exact first-principles universal constant. In that sense, Eq. (1) should be interpreted as the accumulated Fisher-transport burden carried by a split matter-wave configuration, bounded by the same geometric capacity scale that governs gravitational collapse in the macroscopic sector of the framework.

The importance of Eq. (1) is structural: it isolates a kinematic quantity that grows linearly with T and quadratically with momentum separation, but does not by itself yet specify the scale of the coherence-support cell on which that burden is registered.

2.2 Fixed-cell ansatz

The second ingredient is the fixed-cell hypothesis. The operational cell size is taken to be fixed before the interferometric kicks are applied:

$$A_* = \eta\,\lambda_{\rm coh,0}^2, \tag{2}$$

where $\lambda_{\mathrm{coh},0}$ is the pre-split coherence length of the atomic source and η is an order-unity geometric factor. In parallel, the coherence-support area of the interferometric branch is written

$$A_{\mathrm{coh}} = \pi\, w_{\mathrm{coh}}^2, \tag{3}$$

with w_{coh} the transverse coherence-support width relevant to the evolving cloud.

In the absence of a fully microscopic derivation of the operational cell scale, the natural leading phenomenological choice is to tie that scale to the pre-split coherence properties of the atomic source rather than to the later momentum burden imposed by the interferometer sequence. Since those coherence properties are determined at leading order by the thermodynamic preparation of the ensemble, the ansatz $A_* = \eta\, \lambda_{\mathrm{coh},0}^2$ states that the effective coherence cell is set before the LMT sequence begins, while the subsequent kinematic splitting tests whether that fixed cell can support a transport burden growing as $T n^2 k_{\mathrm{eff}}^2$.

This separation of roles is the central phenomenological assumption of the Letter:

> *Thermodynamics fixes the pre-split coherence cell; kinematics stresses that cell through the accumulated transport burden.*

The resulting residual dephasing law then becomes

$$D_{\mathrm{pred}} \equiv -\ln V_{\mathrm{res}} = \frac{\pi\chi}{\eta}\, \mathcal{I}_*/A\, c\, \ell_{\mathrm{P}}\, T\, n^2 k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2. \tag{4}$$

Equation (4) is the master phenomenological equation of the present Letter.

Several features are immediate.

First, the leading dependence is

$$D_{\mathrm{pred}} \propto T\, n^2 k_{\mathrm{eff}}^2. \tag{5}$$

Thus, at fixed T, the residual dephasing scales quadratically with LMT order.

Second, the amplitude of the effect is controlled by the ratio $(w_{\mathrm{coh}}/\lambda_{\mathrm{coh},0})^2$. This makes physical sense: an extended coherence-support region carried by a comparatively small pre-split coherence cell produces a larger effective burden.

Third, the bridge is explicitly phenomenological. The present Letter does not claim that Eq. (2) has already been derived microscopically from the full NDK transport kernel. The claim is more modest and more testable: if the pre-split source coherence sets the operational cell scale, then Eq. (4) is the natural lowest-order forward prediction.

2.3 Baseline benchmark

A representative benchmark is obtained by taking

$$\mathcal{I}_*/A \approx 3.28, \qquad \chi = 1, \qquad \eta = 1, \qquad k_{\mathrm{eff}} = 10^7\,\mathrm{m}^{-1}, \tag{6}$$

together with

$$w_{\mathrm{coh}} = 100\ \mu\mathrm{m}, \qquad \lambda_{\mathrm{coh},0} = 3\ \mu\mathrm{m}. \tag{7}$$

Here $\mathcal{I}_*/A \approx 3.28$ is the representative benchmark value adopted from the cosmological SMBH-envelope calibration of [1] discussed above, and is held fixed throughout the present Letter rather than varied as an independent laboratory parameter.

For a current-tech benchmark $T = 1.4$ s and $n = 100$, Eq. (4) gives

$$D_{\mathrm{pred}} \approx 7.77 \times 10^{-5}, \qquad V_{\mathrm{res}} = e^{-D_{\mathrm{pred}}} \approx 0.99992.$$

Thus the predicted residual dephasing remains far below the percent level in this regime. By contrast, for a MAGIS-like benchmark $T = 3.0$ s and $n = 1000$, the same equation gives

$$D_{\rm pred} \approx 1.66 \times 10^{-2}, \qquad V_{\rm res} = e^{-D_{\rm pred}} \approx 0.9835.$$

This moves the prediction directly into the percent-level contrast-loss regime. In this baseline realization, the theory therefore predicts a clear parametric separation between $n \sim 10^2$ and $n \sim 10^3$ architectures.

3 Parameter sweep

The baseline comparison and the broad parameter sweep are summarized together in figure 1. The plotted median curves illustrate the raw n^2 leverage of the fixed-cell ansatz for representative time scales, while the shaded bands show the spread obtained under a broad variation of coherence and geometric parameters. The key qualitative point is immediate: the $T = 3.0$ s regime is displaced upward relative to the $T = 1.4$ s regime, and the percent-level threshold is reached only in the large-LMT limit.

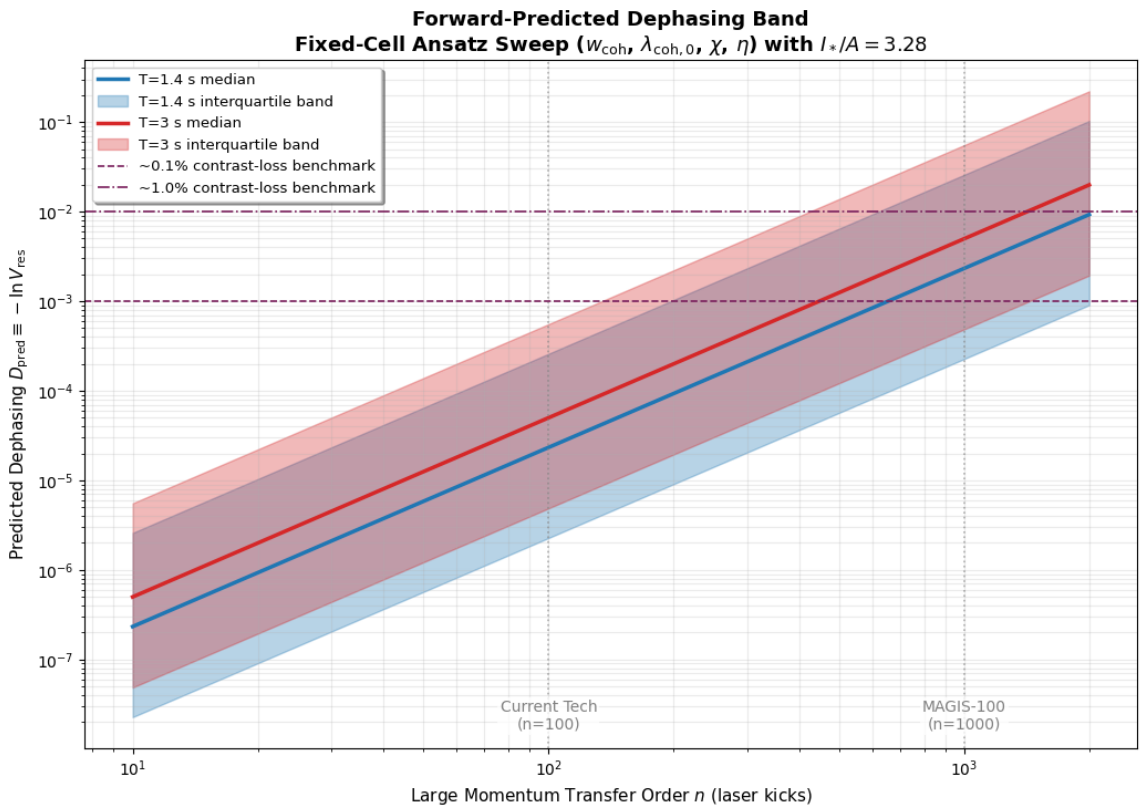

Figure 1: Forward-predicted dephasing band from the fixed-cell ansatz. The solid curves show the median prediction for representative $T = 1.4$ s and $T = 3.0$ s regimes, while the shaded bands show the interquartile spread obtained by sweeping $w_{\rm coh}$, $\lambda_{\rm coh,0}$, χ, and η over broad representative ranges, with the directional-strain threshold held fixed at the adopted benchmark value $\mathcal{I}_*/A \approx 3.28$ inferred from the SMBH-envelope analysis of [1]. The $T = 3.0$ s regime remains systematically above the $T = 1.4$ s regime, while the n^2 scaling is preserved throughout. Horizontal reference lines at 10^{-3} and 10^{-2} indicate sub-percent and percent-level contrast-loss benchmarks.

To test the robustness of that separation, the fixed-cell prediction shown in figure 1 is scanned over

$$w_{\rm coh} \in \{10, 30, 100\}\,\mu{\rm m}, \quad \lambda_{\rm coh,0} \in \{0.3, 1, 3, 10\}\,\mu{\rm m},$$
$$\chi \in \{0.1, 1, 10\}, \quad \eta \in \{0.3, 1, 3\},$$

while the directional-strain threshold is held fixed at the adopted benchmark value $\mathcal{I}_*/A \approx 3.28$ inferred from the SMBH-envelope calibration of [1]. The shaded band shows the interquartile spread of forward predictions for each T.

Although four nuisance quantities are scanned, the normalization of Eq. (4) depends only on the composite combinations

$$\frac{\chi}{\eta} \qquad \text{and} \qquad \left(\frac{w_{\rm coh}}{\lambda_{\rm coh,0}}\right)^2.$$

The sweep should therefore be understood not as introducing four independent degrees of freedom in the final prediction, but as sampling a representative range of the two effective geometric/coherence ratios that control the overall dephasing amplitude.

For clarity, the quantities scanned here are not additional free parameters of the underlying NDK threshold law itself. The imported directional-strain benchmark $\mathcal{I}_*/A$ is held fixed throughout at the cosmologically calibrated value discussed above. The sweep instead samples uncertainty in the source/coherence properties of the interferometer $(w_{\rm coh}, \lambda_{\rm coh,0})$ together with residual order-unity geometric factors (χ, η) entering the fixed-cell bridge from the substrate prediction to a concrete experimental architecture. In that sense, the shaded band should be read as a phenomenological transfer band for the laboratory implementation, rather than as a re-fitting of the underlying NDK threshold.

The baseline medians and the parameter sweep both support the same qualitative conclusion. The fixed-cell ansatz does not predict a universal constant dephasing floor, but rather a residual geometric burden that grows as $Tn^2k_{\rm eff}^2$, with current $n \sim 10^2$ devices expected to remain significantly below the dephasing regime accessible to future $n \sim 10^3$ architectures.

This is precisely why the LMT-order scan is the crucial experimental discriminator. Technical noise may have many apparatus-specific dependencies, but the NDK residual forecast singles out a simple leading structure:

$$D_{\rm pred}(n) \;=\; a_0 + a_1 n^2 \qquad \text{at fixed } T, \tag{8}$$

with the fitted slope itself growing linearly with T. More explicitly, the experimental discriminator is not the mere appearance of additional contrast loss, but the emergence—after standard technical-noise and environmental modeling—of a residual law whose leading dependence is quadratic in n at fixed T, and whose fitted quadratic coefficient scales linearly with T across runs with different interrogation times. Within the fixed-cell ansatz, the normalization of this residual law is then compared against the benchmark band shown in figure 1, with $\mathcal{I}_*/A$ held fixed at the cosmologically imported value rather than re-fit from laboratory data. If such a residual pattern appears, the experiment will not merely have seen unexplained contrast loss; it will have isolated the specific n^2 geometry predicted by the fixed-cell bridge.

Conversely, if future LMT interferometers operating in the relevant T, n, and coherence regime fail—after standard technical-noise and environmental modeling—to exhibit a residual contrast-loss law whose leading dependence is quadratic in n at fixed T, with the fitted quadratic coefficient scaling linearly with T, then the fixed-cell bridge proposed here is falsified.

4 Conclusion

This Letter has presented a compact phenomenological bridge between the NDK transport framework and LMT atom interferometry. The bridge is built from a dimensionless transport-burden variable and a fixed pre-split coherence-cell ansatz. Together they yield a forward residual-dephasing prediction,

$$D_{\mathrm{pred}} = \frac{\pi\chi}{\eta}\,\mathcal{I}_*/A\,c\,\ell_{\mathrm{P}}\,T\,n^2 k_{\mathrm{eff}}^2 \left(\frac{w_{\mathrm{coh}}}{\lambda_{\mathrm{coh},0}}\right)^2,$$

whose defining experimental feature is a leading quadratic dependence on LMT order at fixed interrogation time.

Two conclusions follow.

First, the fixed-cell ansatz establishes a strict physical separation between the thermodynamic preparation of the atom and its subsequent kinematic stress. This is what protects the n^2 signature and makes the proposal experimentally meaningful.

Second, the combination of a baseline benchmark and a broad sweep over coherence and geometric parameters, with $\mathcal{I}_*/A$ held fixed at the adopted cosmological benchmark value ≈ 3.28, indicates a clean regime separation: current $n \sim 10^2$ instruments remain parametrically below the percent-level window, whereas $n \sim 10^3$ architectures of the MAGIS-like type enter the 1% to few-% residual-dephasing regime for a physically plausible range of coherence assumptions. In that sense, the present framework makes a sharp operational forecast:

> *If the fixed-cell ansatz is the correct bridge between NDK transport and macroscopic atom interferometry, then future $n \sim 10^3$ LMT interferometers should observe a residual contrast loss at the percent level whose leading dependence is quadratic in LMT order.*

That forecast is sharp enough to be falsifiable. If LMT atom interferometers operating in the relevant T, n, and coherence regime do not reveal a residual contrast-loss law whose leading dependence is quadratic in n at fixed T, with the quadratic coefficient scaling linearly with T, then the fixed-cell bridge developed here is falsified as the proposed laboratory realization of NDK transport.

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